

FALCON's Response to Amazon for Supply of Linear CBS 8K PPH

September 20, 2023



www.falconautotech.com



Kind Attention –

Mr. Rahul Didwani

Ms. Akansha Jain

M/s Amazon

Offer Ref: F23-00430; Date: 20-09-2023

Subject – Techno -Commercial Offer of Linear Cross Belt Sorter 8K PPH

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

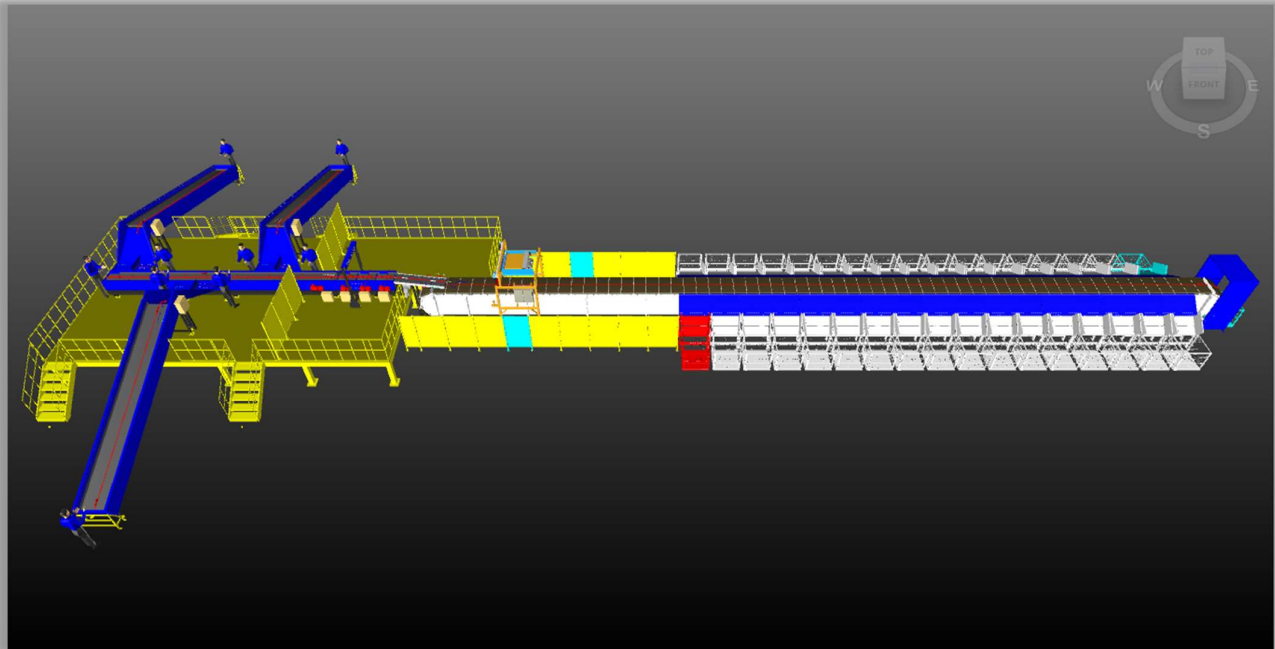
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFP for Linear Cross Belt Sorter 8K PPH



Proposal Reference – F23-00430

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1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Shipments Per Hour
3	ICR	Intelligent Character Recognition
4	MEZZ	Mezzanine
5	LIM	Linear Induction Motors
6	AC	Alternating Current
7	DC	Direct Current
8	PLC	Programmable Logic Controller
9	IT	Information Technology
10	BOQ	Bill Of Quantity
11	I/O	Input/ Output
12	PDP	Power Distribution Panel
13	PC	Personal Computer
14	UPS	Uninterrupted Power Supply
15	CBS	Cross Belt Sorter
16	MDR	Motor Driven Roller
17	DAP	Design Approval Phase

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this Amazon RFQ of Linear CBS. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success. Following the same objective for the CBS system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price. Our solution is based on the following key characteristics:

- **Linear CBS for shipment sorting, employing cross belt technology, have been designed to handle a throughput of 8,000 packages per hour each.**
- **This Linear CBS is equipped with infeed conveyor system, comprising of single Top Loader Induct.**
- **In the system, there are 32 Primary chutes (Sliding Type), 1 rejection chute & 1 Sort fail chute.**
- **The system layout has been meticulously planned to facilitate smooth operational flow, ensuring efficient movement of personnel.**

A tailor-made and simple layout designed explicitly for Amazon. The proposed layout is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our meeting.

1. **Falcon's reliable Shipment sortation systems**

These systems are globally being used by most innovative brands such as Amazon, Amazon Delhivery, Asendia, Fastway and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Amazon satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

3. **A strong and already successful partnership between Amazon & Falcon**

Amazon has been Falcon's customer since long and using multiple Falcon Systems. For each project, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, regular governance meetings and monthly reports.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5
Product Lines

15+
Countries with Live Installations

1800+
Total Installed Systems Globally

1Million+
Sorts Per Hour

600+
Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

تجارت

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Handelman Online Limited

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING

Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

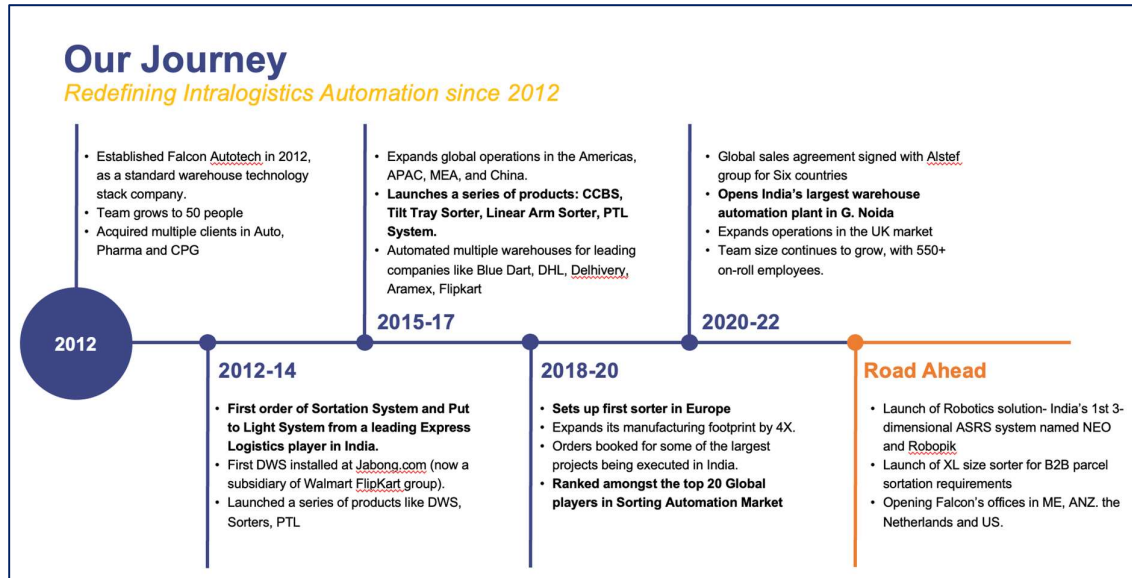
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
 - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
 - Mini (600MM)
 - Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

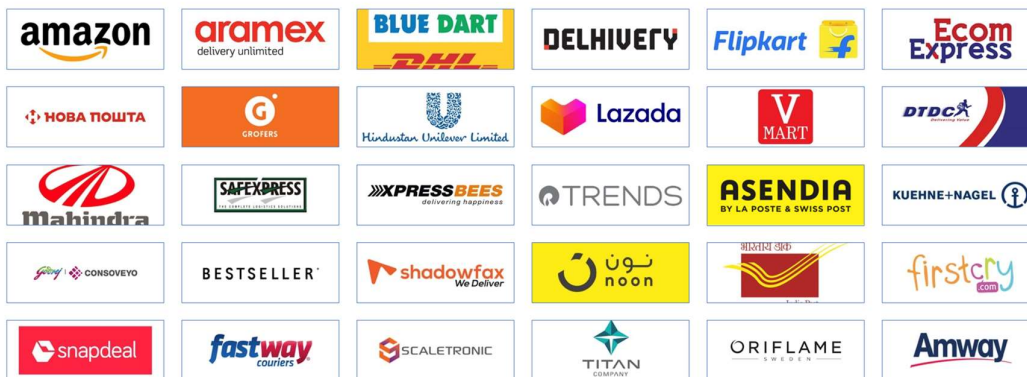


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

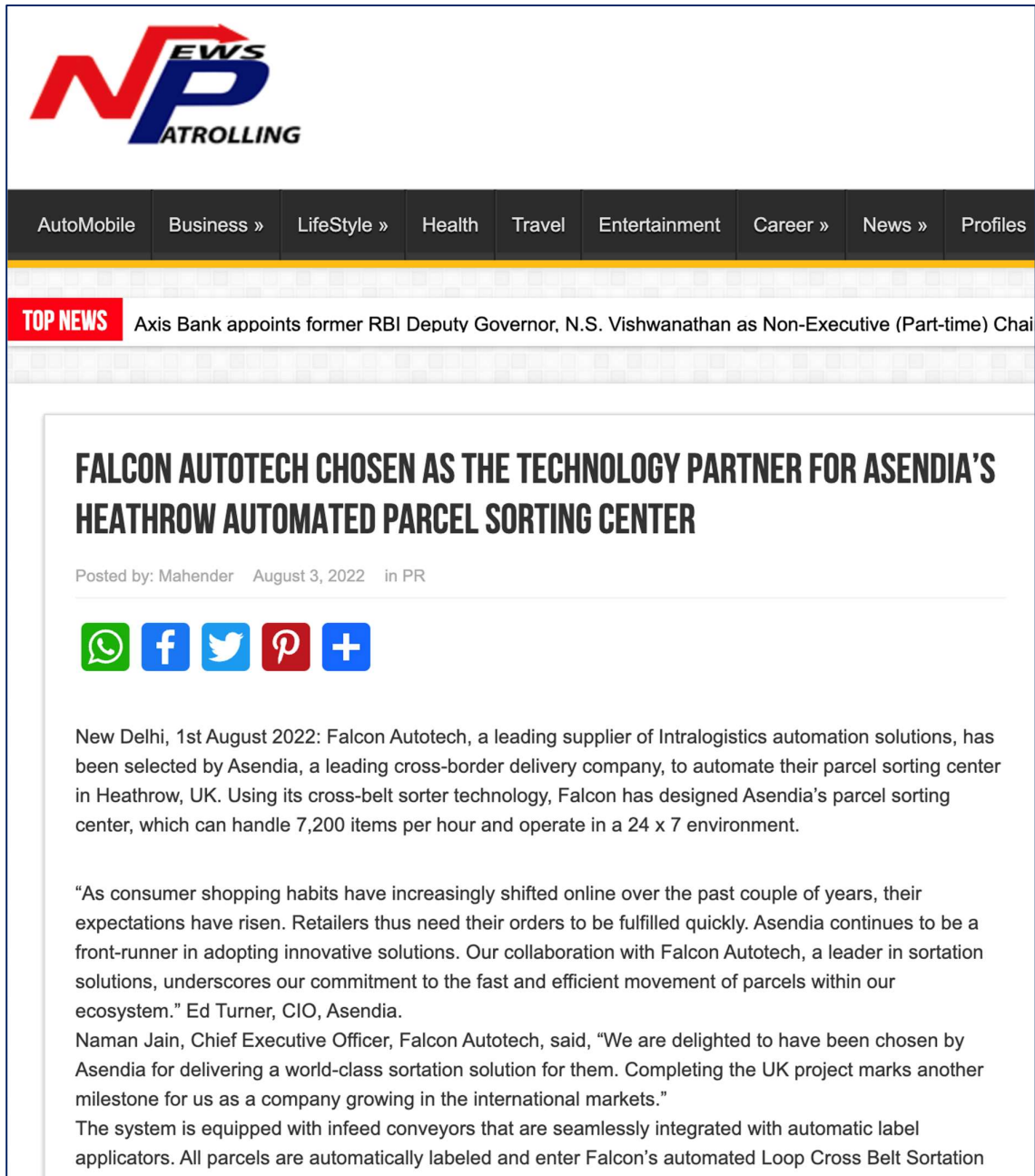


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Linear Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of shipments arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Shipment handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Shipment, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 shipments per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Linear Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

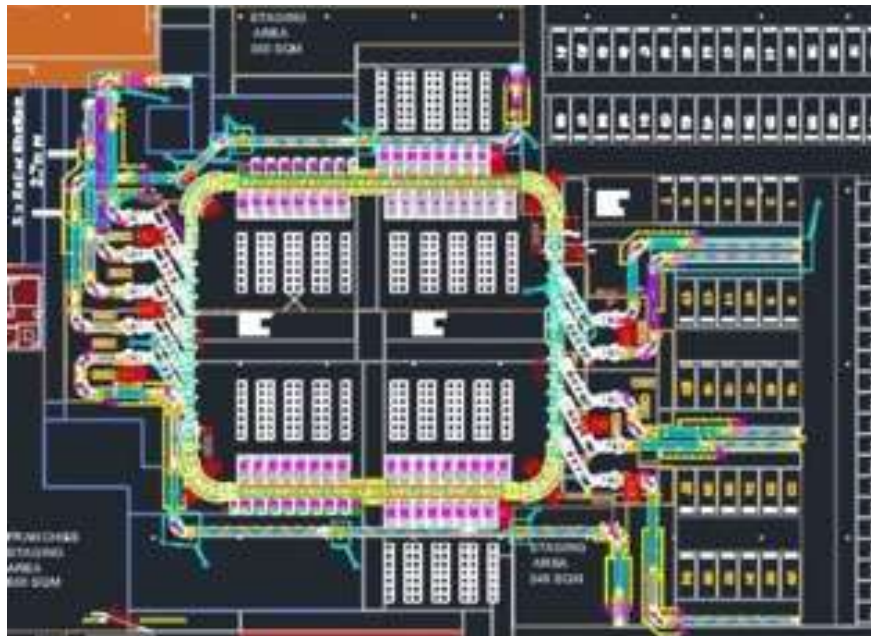
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

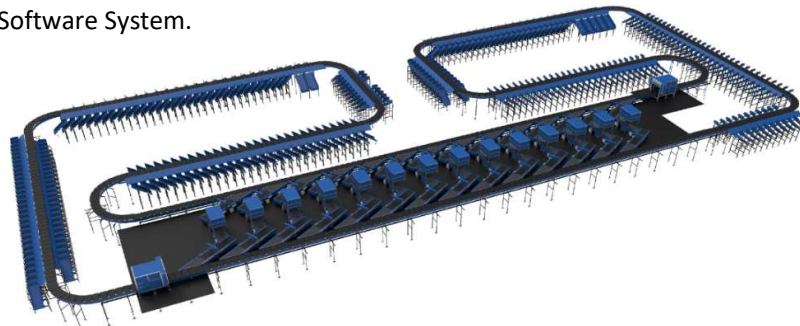
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

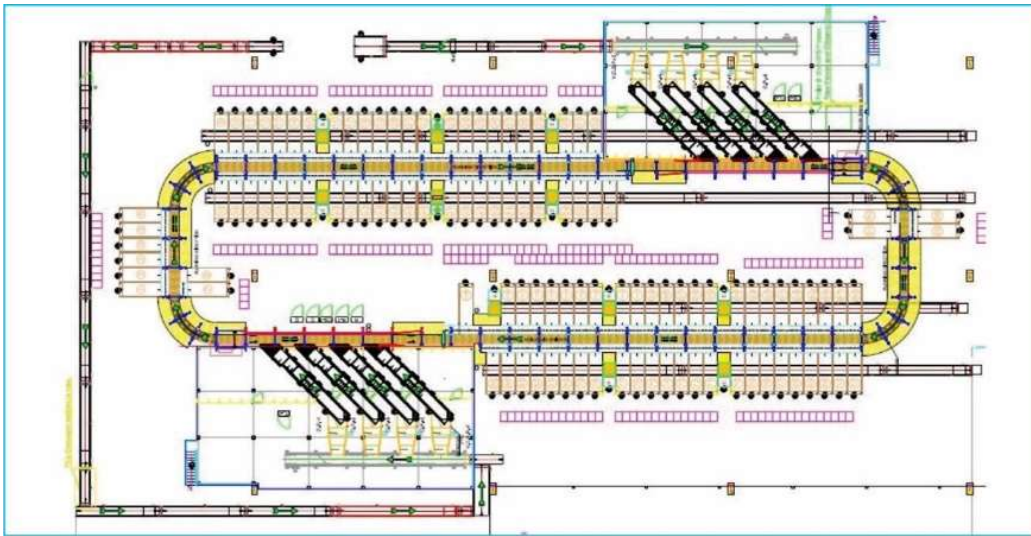
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Linear Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detailed.

Falcon purposes to use its “Dual Belt Cross Belt Sorter” to provide the maximum benefits to Amazon in-terms of handling large shipment sizes and weight.

6.1 Shipment size loadable on the sorter

Falcon's Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	400
Max Width	mm	400
Max Height	mm	400
Max Weight	Kg	3
Min length	mm	100
Min Width	mm	100
Min Height	mm	50
Min Weight	gm	200

6.2 Shipments to be loaded on Sorter shall have the following characteristics:

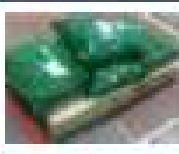
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Sortable Shipments

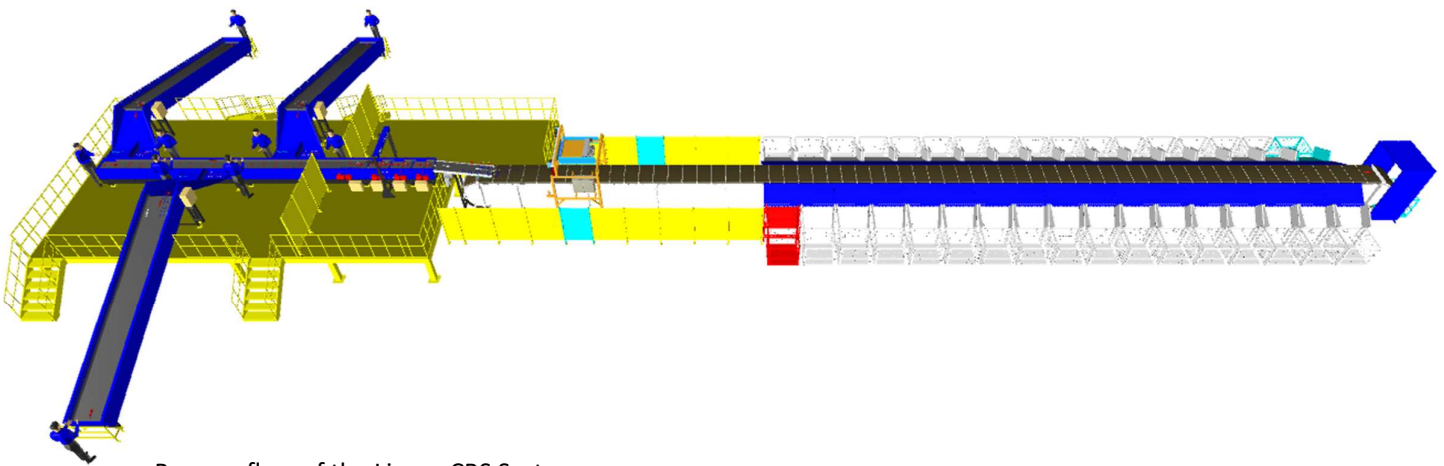
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (30 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Linear CBS for sorting shipment as per Amazon's logic.

7.2 Summary of the System



Process flow of the Linear CBS System-

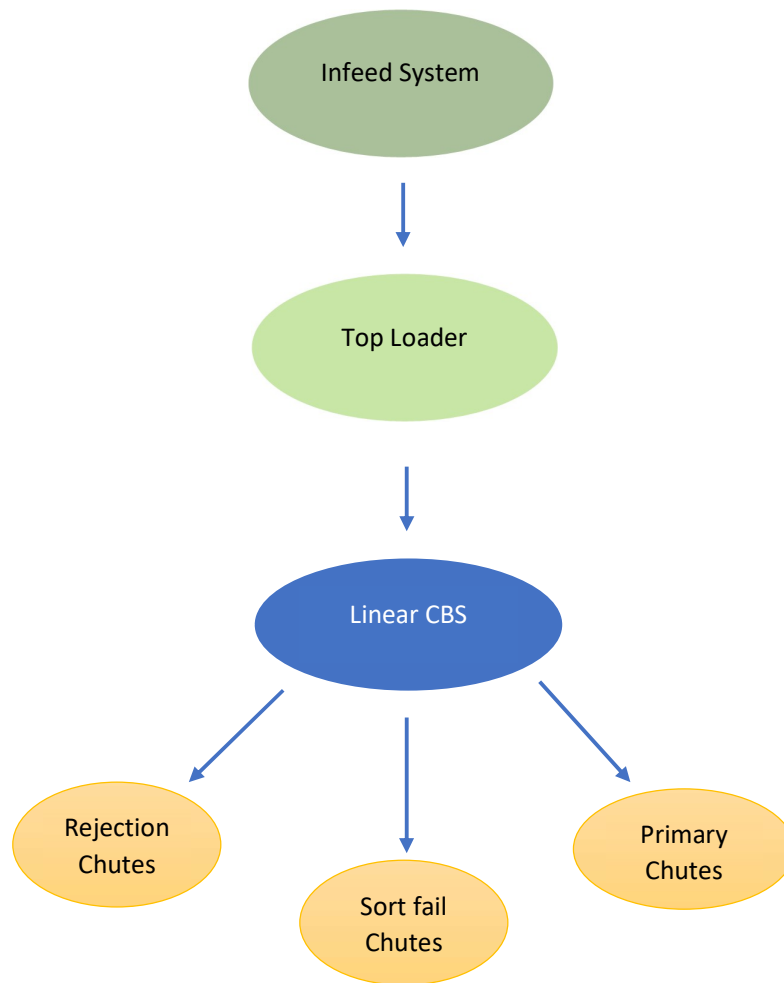
1. **Infeed System:** - The shipments are dumped in bulk onto the Launch Pad, which then convey the shipments from a lower level (below the mezzanine) to Induction line.
2. **Inducts:** - After the shipments are collected in the infeed line, an operator picks and positions each shipment on the induct line, ensuring that the shipment is properly aligned and that its barcode is facing upwards. The feedlines then automatically induct the shipments onto the Linear Cross-Belt Sorter.
3. **Linear CBS:** - Once the shipments have entered the main Linear CBS, Cross-Belt Sorter (CBS) efficiently sorts the shipments into their respective output chutes by utilizing the data provided by Amazon's sorting logic.
4. **Output Chutes:** - The shipments are discharged output chutes.
 - a. **Primary Chute (Sliding-type)** - Within a Linear CBS system, there are a total of 32 sliding-type chutes.
 - b. **Rejection Chute-** One rejection chute is present to handle rejected Shipments.
 - c. **Sort fail Chute-** Two Sort fail Chutes are present to collect shipment that missed to get sorted in either of the chutes.

7.3 Main benefits of the proposed solution

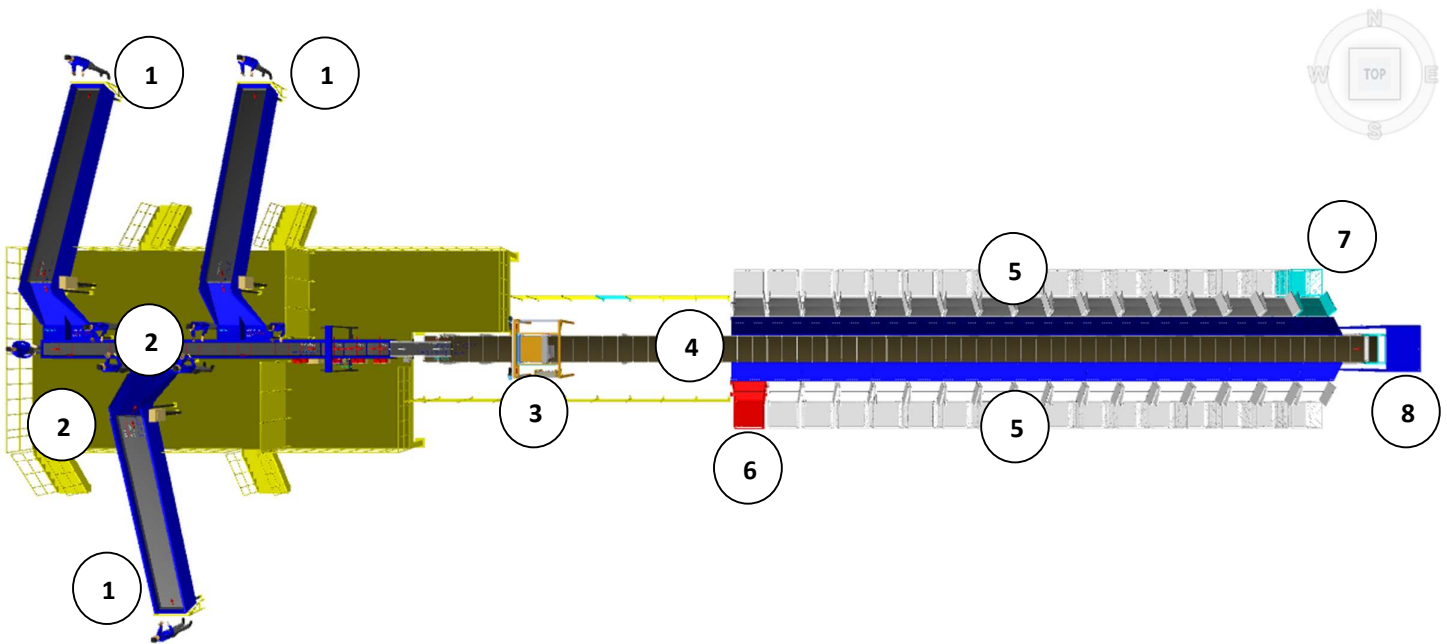
The proposed solution has the following advantages:

1. High operational throughput
2. Good Shipments conveying quality for the entire range of conveyable Shipments with a sorting conveyor speed of up to 1.4 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for the increased number of splits in limited space
5. FALCON's CBS can adapt to changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Concept Description



9. Layout View

**Legend**

- 1 – Infeed Lines
- 2 – Induct lines
- 3 – ICR Tunnel
- 4 – Main Linear CBS
- 6 – Primary chutes (Sliding-type)
- 7 – Rejection chute (red colour)
- 8 – Sort fail chute

10. Proposed System Capacity Calculations

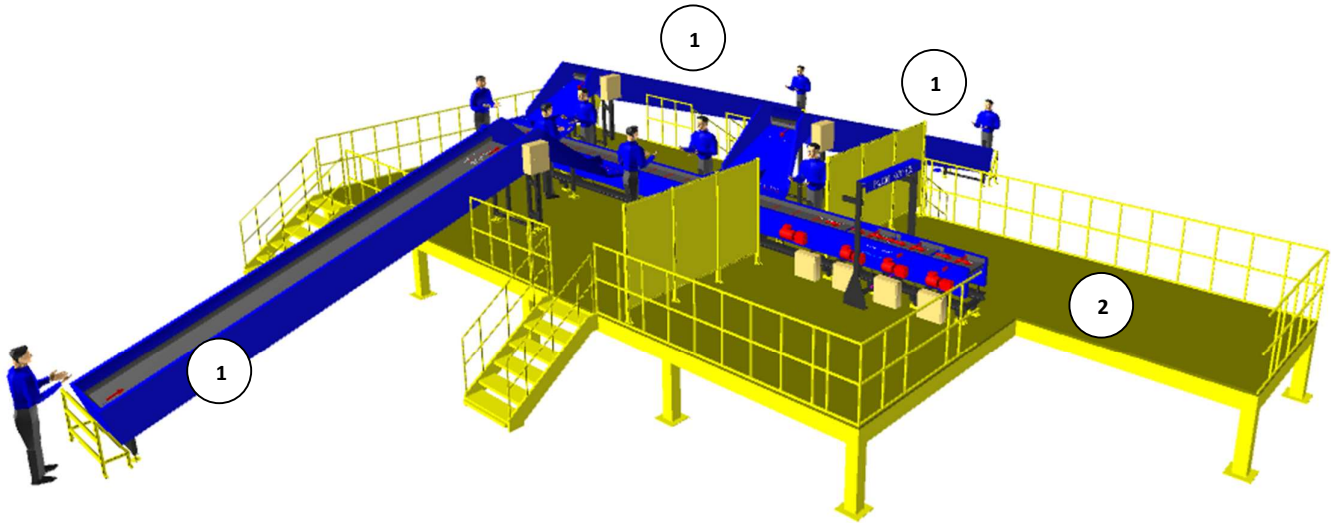
10.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on Amazon requirements.

SPECIFICATION	VALUE
Sorter Type	Linear CBS
Speed	1.4 m/s
Pitch	500 mm
Carrier per hour	10,080 CPH
Belts per hour	10,080 BPH
Designed Throughput of 1 Linear CBS (A)	10,080 Shipments per hour
Per Feedline Capacity	8,000 Shipments per hour
No of Feedlines	01 Nos.
Total Feedline designed capacity (B)	8000 Shipments per hour
System designed throughput (Lower of A&B)	8000 Shipments per hour

11. System description

11.1 Infeed System



The system is equipped with three infeed conveyors, facilitating the movement of shipments from the lower level to the mezzanine level.

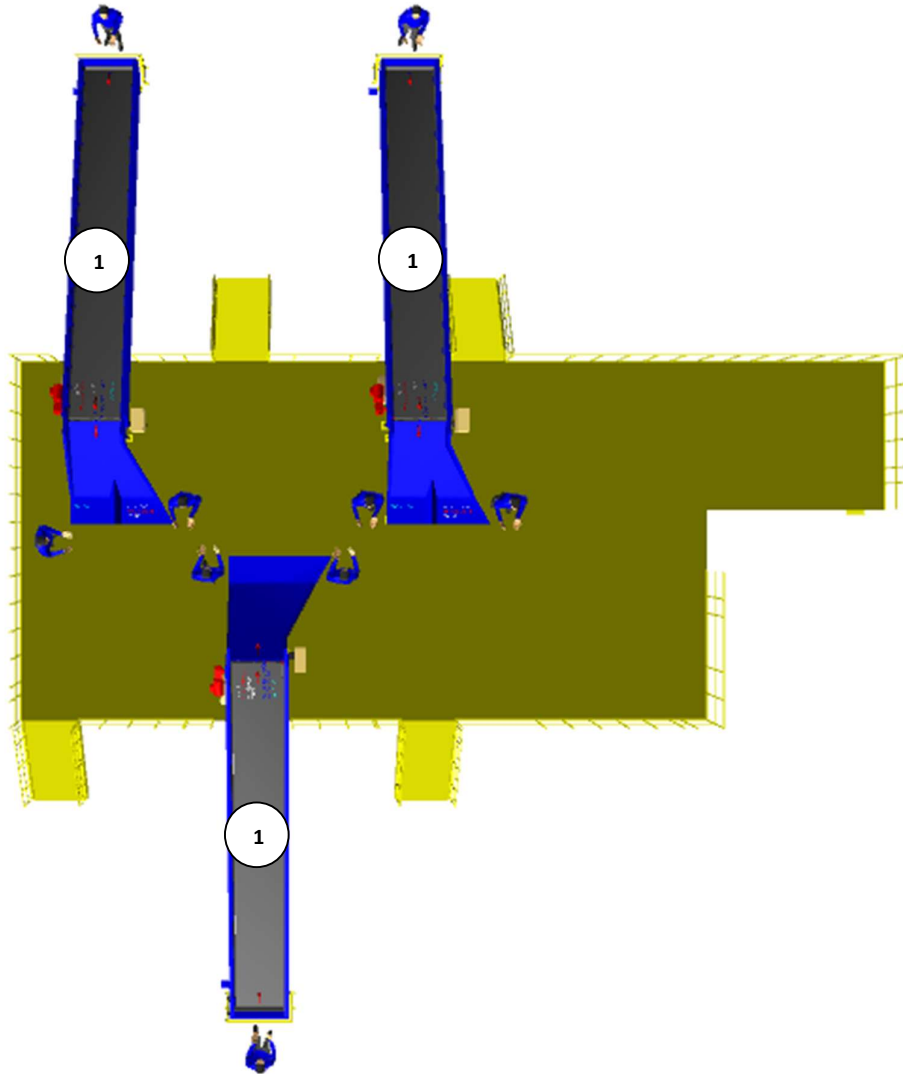
Legend

- 1- Infeed Lines
- 2- Mezzanine

11.1.1 Infeed System

Infeed conveyor system consists of below components:

1. Inclined PVC Conveyor
2. Collection Chutes



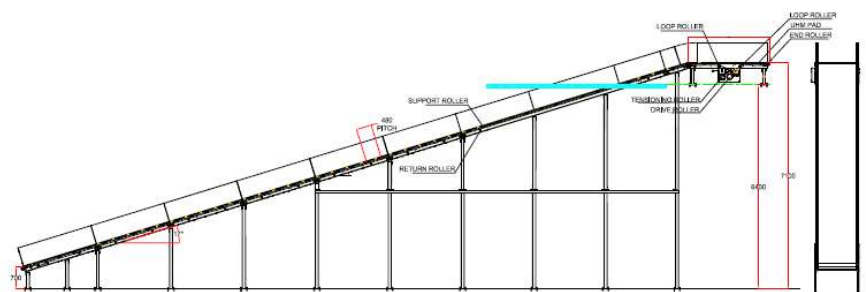
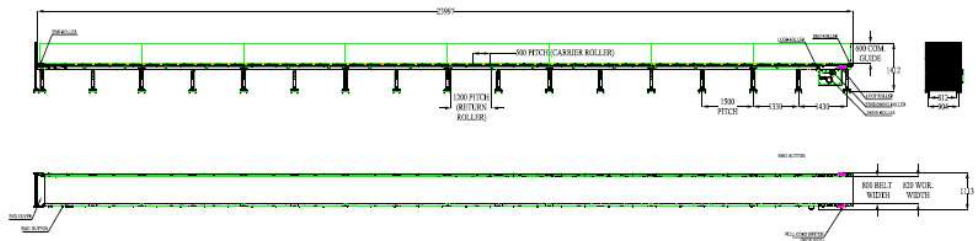
11.1.1.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

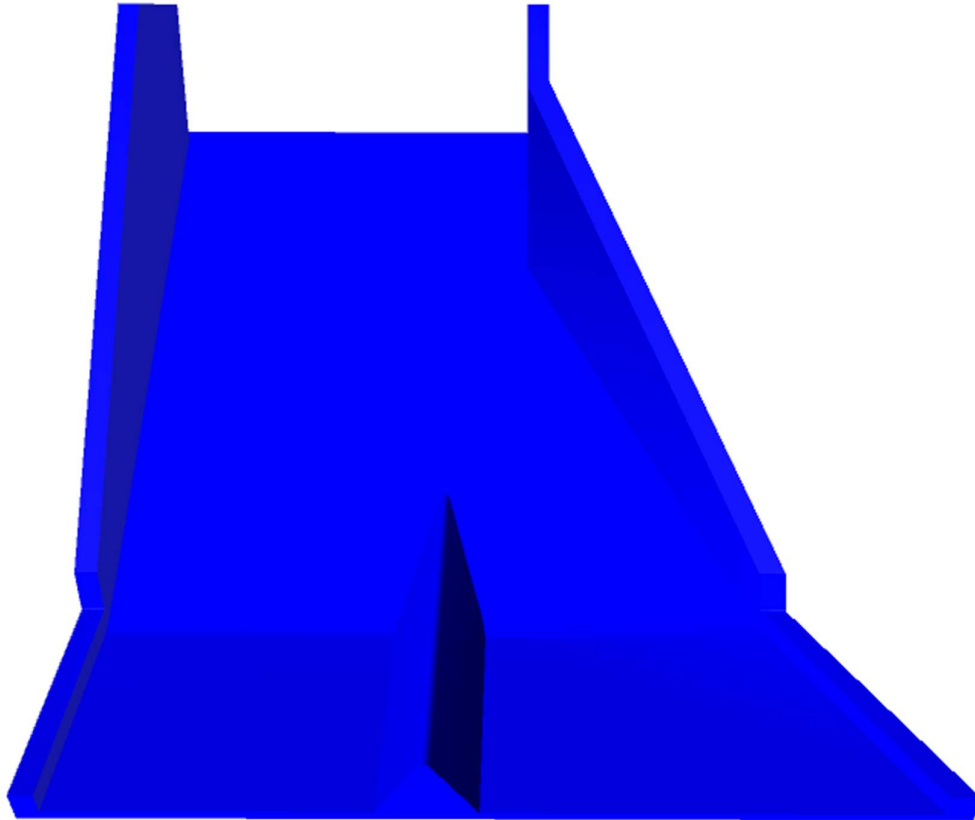
- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

11.1.1.2 Collection Chutes

A collection chute is a type of chute used for the smooth descent & collection of bulk shipment. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area.



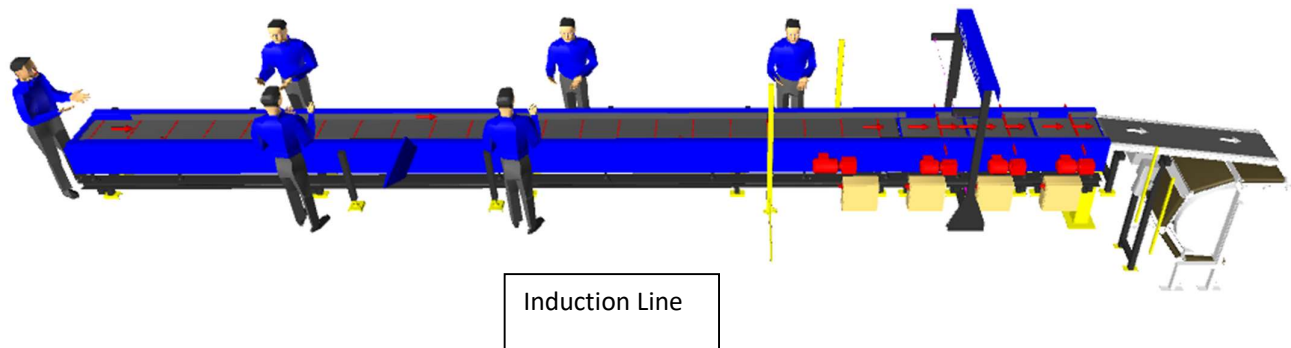
11.1.2 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	Set
1	Launchpad	622	2800	7.2	3
2	PVC Conveyor Inclined	2061	9252	9.3	1

11.2 Induction to Sorter

The main purpose of the feedlines is to transport the shipments from the infeed points to the sorter by means of straight and curved belt conveyors,

The design of the system ensures that the maximum capacity of each feed, considered as the sum of the flows from the connected inductions, is always less than the planned induction capacity of 8000 shipments/hour.



11.2.1 Feedlines

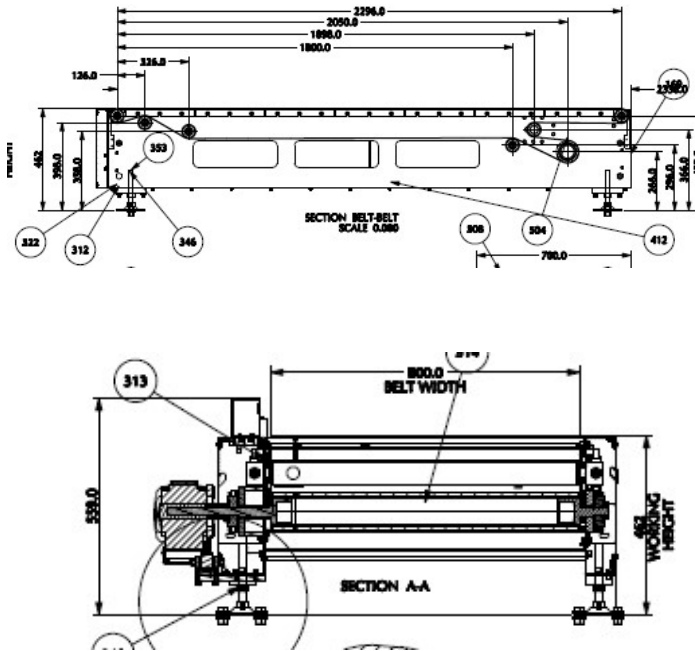
Linear CBS has one Automatic Feedlines with Designed IPP of 8000 PPH.

1. Feedline has Two Conveyor Modules-
 - Loading Conveyor- 1 Nos.
 - Buffer Conveyor- 3 Nos.
 - Top loader Conveyor- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, shipments are evenly spaced and smoothly transferred to the Linear CBS.

11.2.1.1 Buffer Conveyors

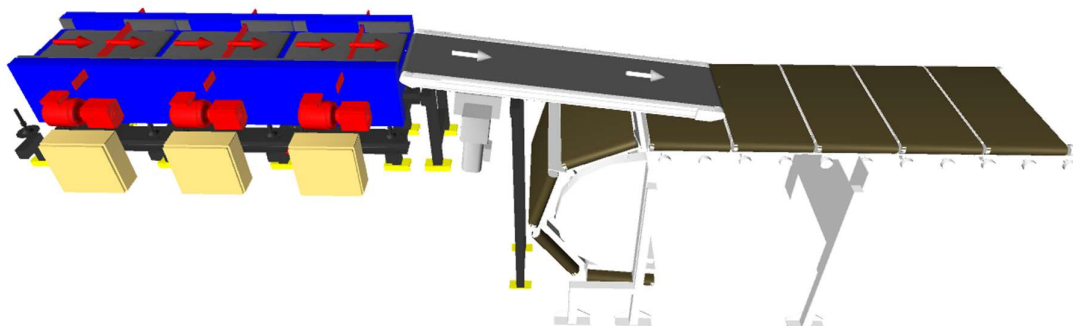
A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

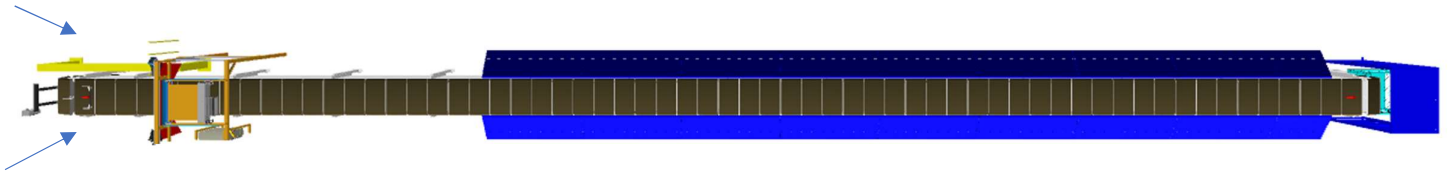
This Conveyors required to maintain the Throughput of Line.



11.2.1.2 Top Loader

A Top Loader is a conveyor that transfer the shipments from Induct line to Main CBS smoothly.



11.2.2 Linear Sorter

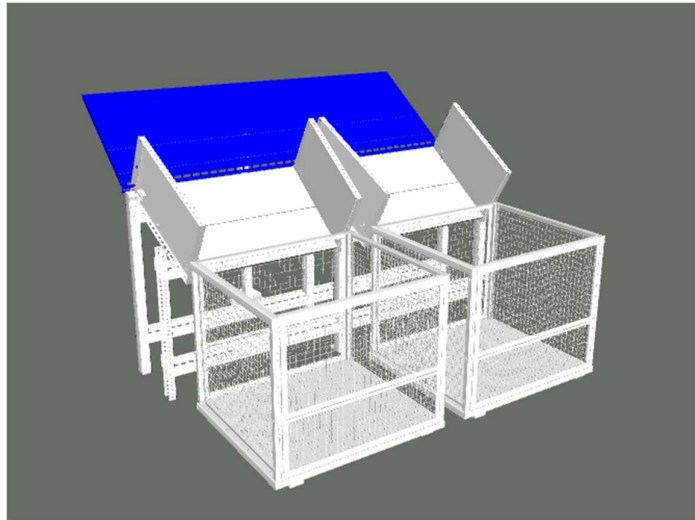
Induct Zone

1. The proposed Single Deck CBS Sorter will be installed on ground level. The Sorter runs at 1700 mm from ground level.
2. We have provided Linear with single belt Carriers with 500 mm pitch and Belt Size of 495 x 800mm.
3. One Linear CBS has a total length of 31 m (Top Side Only) & 131 number of carriers.
4. As Shipments pass through the Barcode Scanning Tunnel on the Linear CBS, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

11.2.3 Sorting Output

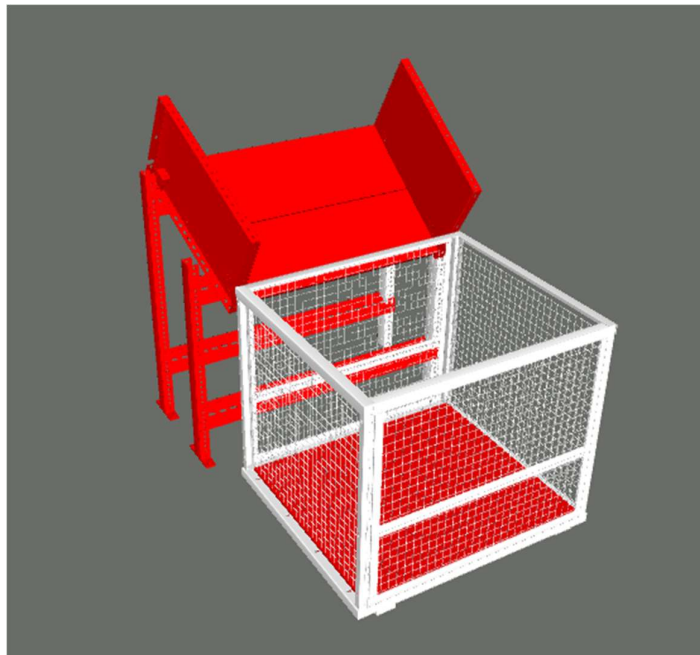
11.2.3.1 Sliding type chute (Primary Chute)

A sliding chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes the trolley placed below to collect the shipments within its area.



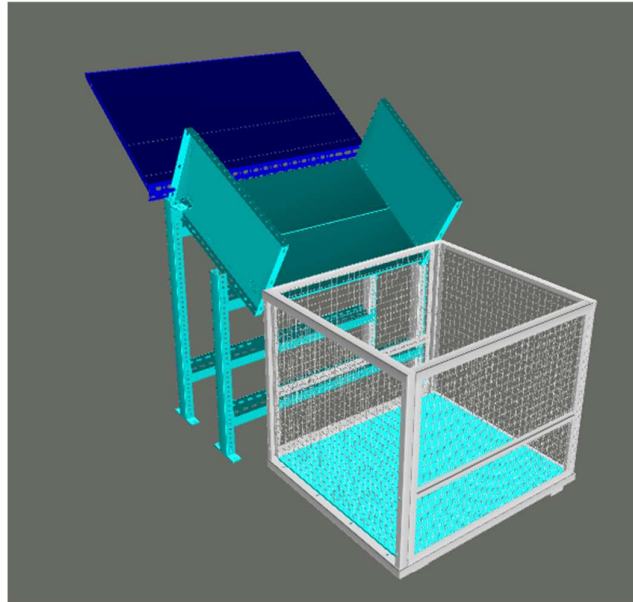
11.2.3.2 Sliding Type Chute (Rejection Chute)

A sliding chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes the trolley placed below to collect the shipments within its area.



11.2.3.3 Sliding Type Chute (Sortfail Chute)

A sliding chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes the trolley placed below to collect the shipments within its area.



Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12. Description of Components of Equipment

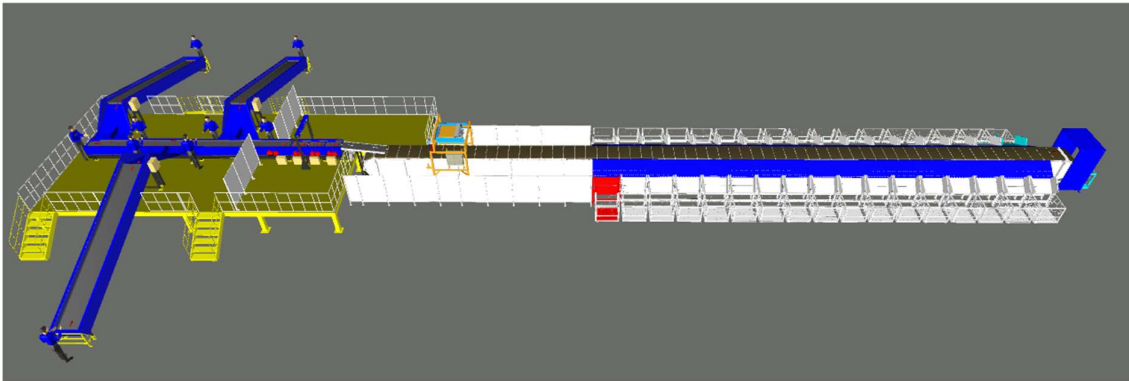
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

12.1 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



12.1.1 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable shipment recirculation.



12.1.2 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



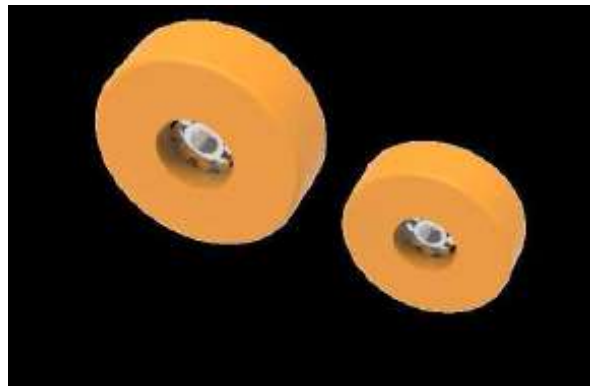
12.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



12.1.4 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



12.1.5 Non-contact based linear motor drive

Extremely High Energy Efficient Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



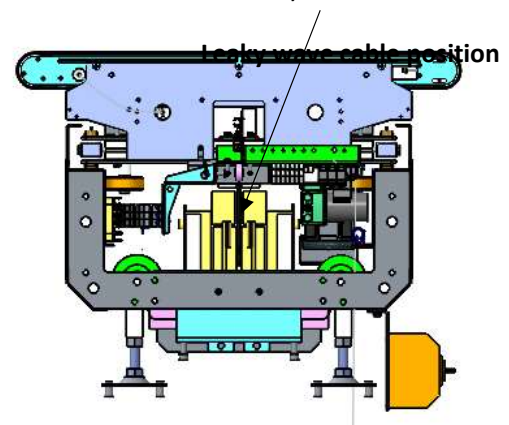
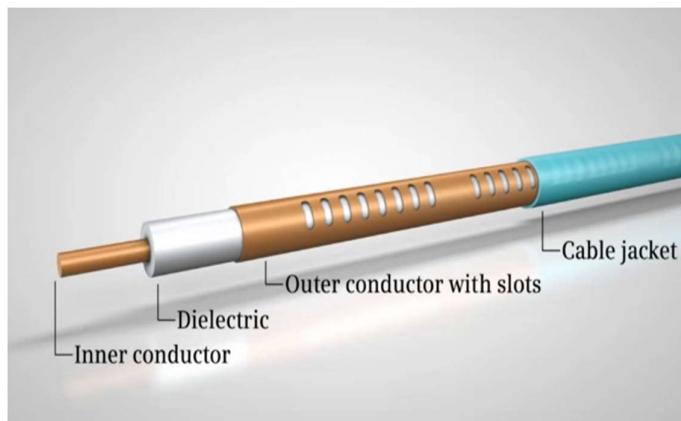
12.1.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



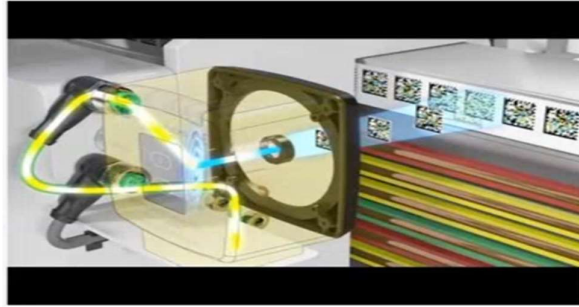
12.1.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



12.1.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the Linear at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter Linear, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



12.1.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Linear CBS
Sorter Speed in m/s	Up to 1.4 m/s
Sorter Linear Length (Top Side Only)	~31 m
Sorter Actuation Technology	Electric
Chutes	
Collection Chute	32
Rejection Chute	1
Sort fail Chute	2
Height of Sorter	~1700 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	500
No. of Carriers	131 Nos.
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Feedlines (Semi-Automatic)	01 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

12.2 Scanning & Sensing on Linear Sorter

12.2.1 Automatic Barcode Scanning

Top Sided Barcode Scanner to scan shipment 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per AMAZON requirements, Scanning is in AMAZON scope.



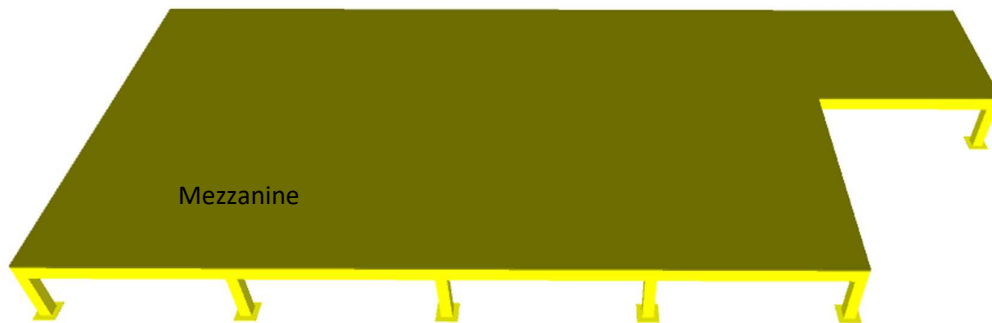
12.2.2 Shipment Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple shipment positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Shipment on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



12.3 Steel Works – Mezzanine & Staircases-

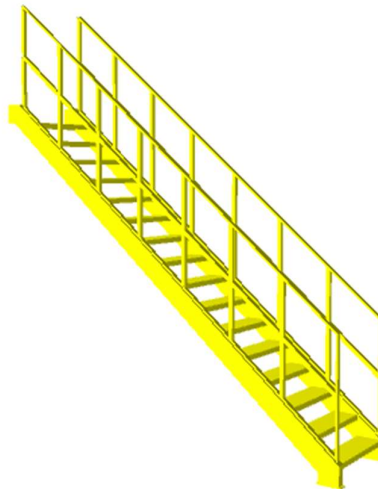
12.3.1 Mezzanine



Plan view of Mezzanine

- Total Mezzanine Area: 96 Sq. Metres
- Clear height below Mezzanine: 1100 mm
- No of Stairs: 4 Nos

12.3.2 Staircase



- No of Stairs: 04 Nos

13. Proposed System Technical Details

13.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	Infeed System - Powered Belt Conveyors - Launchpad Conveyor - Collection Chutes	~9.3 metres (01 Modules) ~7.2 metres (03 Modules) 03 Nos
2	1	Feedlines Consists of - Loading Conveyor - Buffer Conveyor - Top Loader Conveyor	1 Set 3 Set 1 Set
3	1	Sorter 1 Linear Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Product Centring System - Fencing	1700mm (from mezz level) ~31 m (Top Side) ~1.4 m/s Linear Motor 500 mm
4	1	Sorter Outputs - Primary Chute - Rejection Chute - Sort fail Chute	32 Nos 01 Nos 02 Nos
5	2	Steel Works - Operator and Maintenance Platform - Stairs	~96 SQM 04 Nos

13.2 Electrical Equipment

Electricals			
1	1	Consists of Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

13.3 Control System

Components			
1	1	Consists of Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

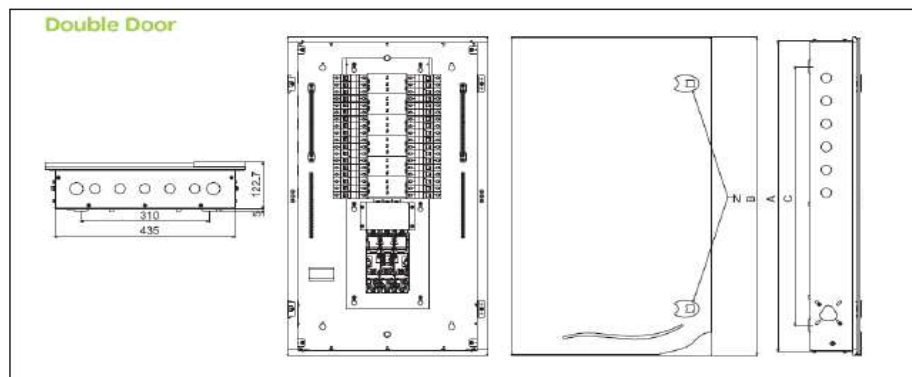
14. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

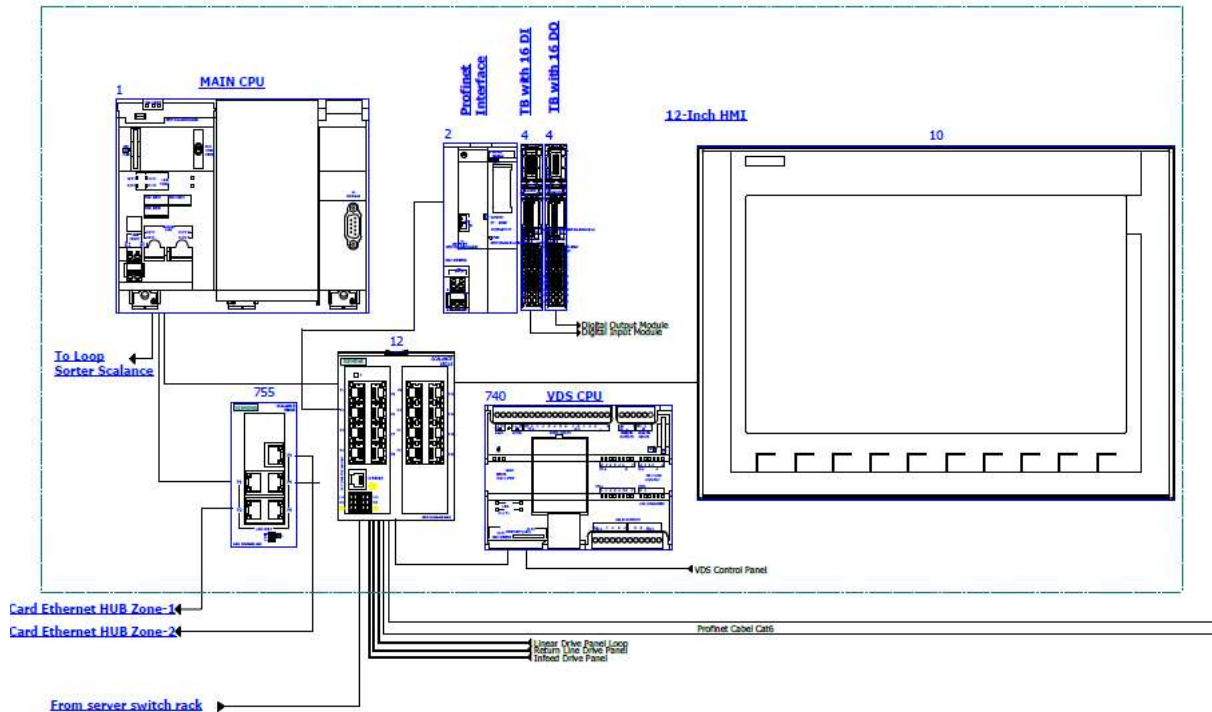
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

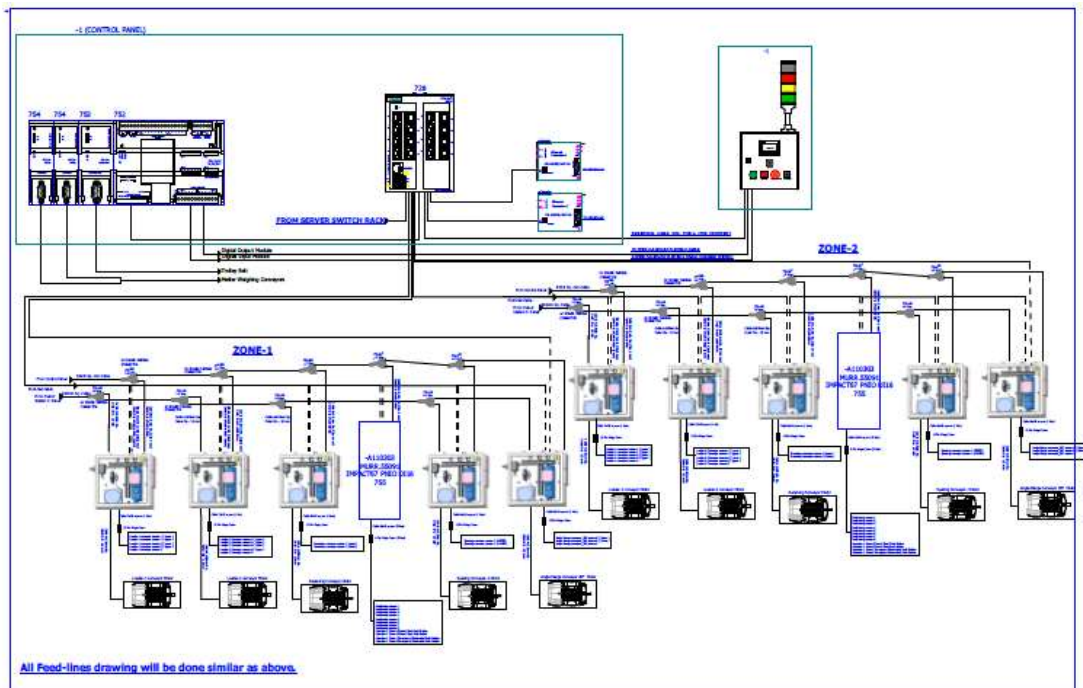
14.1 Reference Picture of Power Distribution Panel



14.2 Main Control Panel (Reference)



14.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

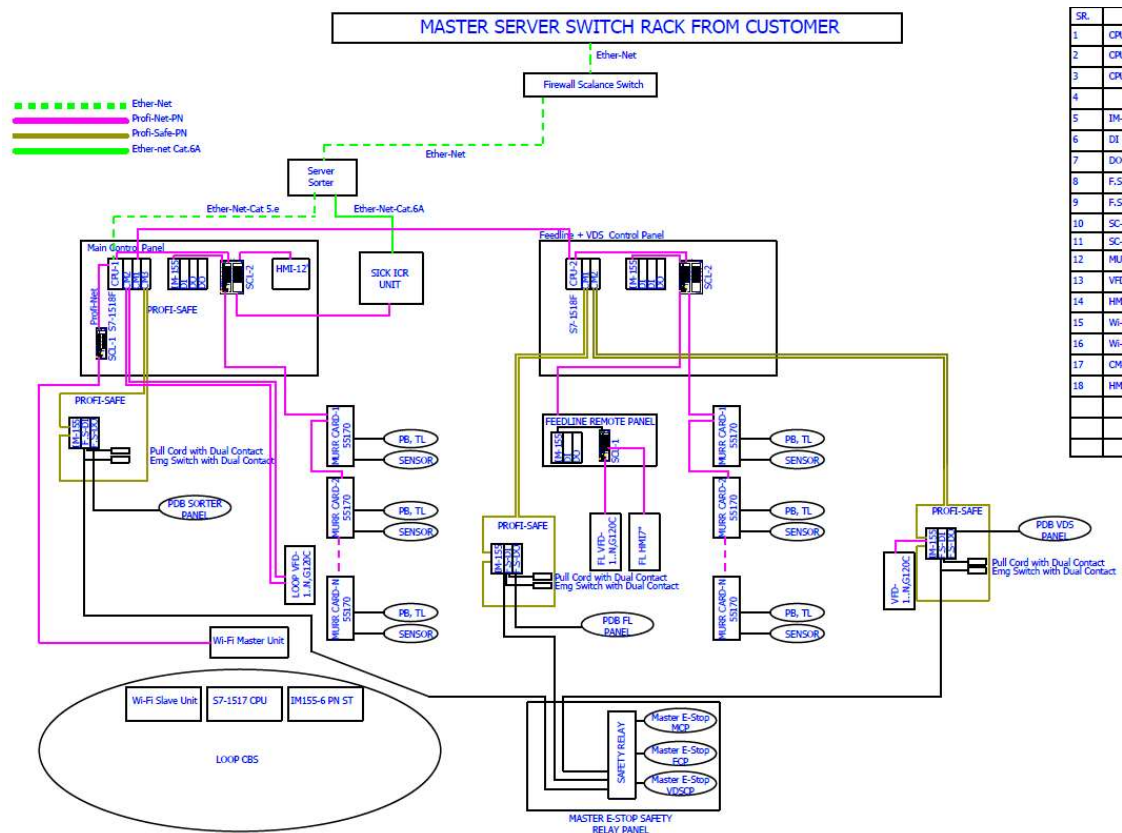
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Linear), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

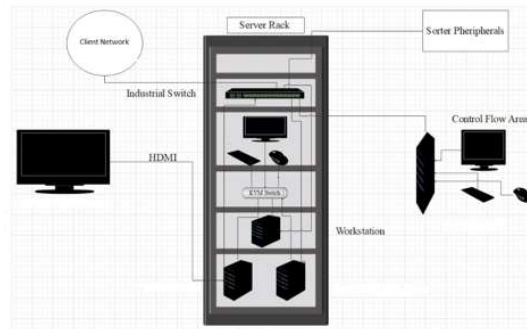
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

15. Controls Architecture (Ref.)



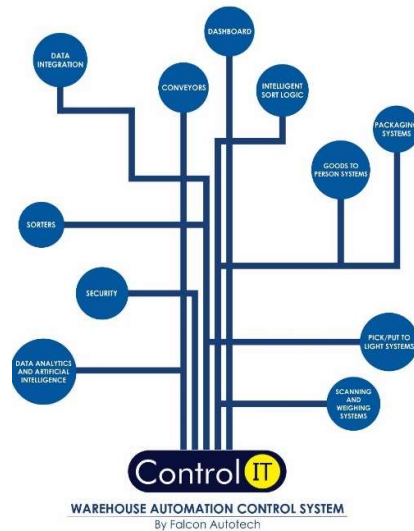
16. IT Architecture (Ref.)



16.1HAA Server (In Amazon Scope)

S.no	Description	Qty
20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40		
1	Synology_storage_DS723+ 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,	2
	2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	
2	Dell Tower Model T440 -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in	2
	Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	
3	Dell PowerEdge T40 Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3	1
	Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

17.Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

18.Falcon's Visual Inspection System (SCADA)

SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities

The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for the Amazon Riyadh Hub. This SCADA system receives from each monitored sub-system all information on their operating status in real time.

At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

19.Key Components Make

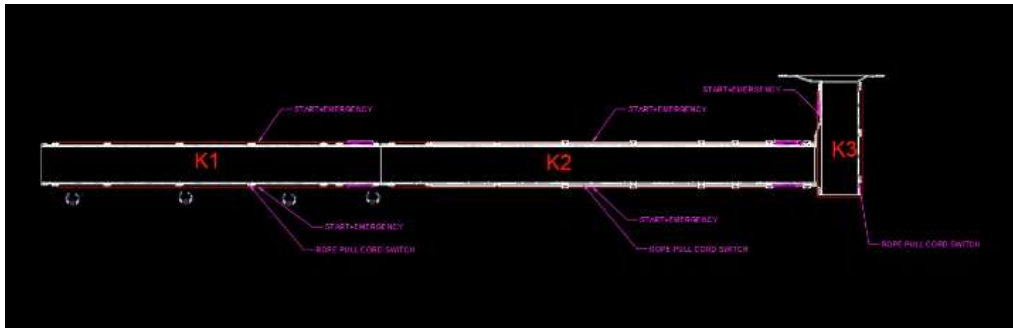
Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Power Transmission Systems	Vahle
HMIs	Siemens

20.Principle of Safety

20.1E-Stops

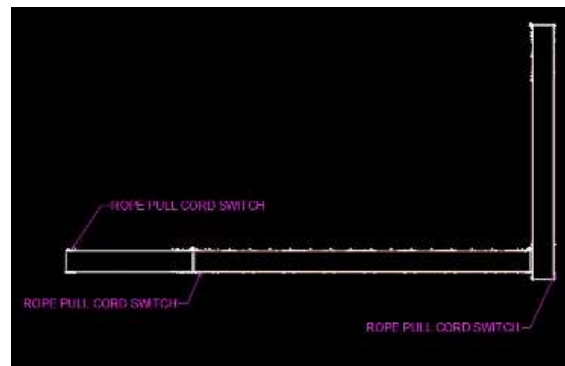


- a. At Every Conveyor Module – Both sides



20.2Pull Cords Switch

Required for Infeed Conveyors



21. Program Organisation

21.1 Program/ Project Schedule

S.No.	Activity	Timeline
1.	Receipt of PO	Week 0
2.	Project Kick off	Week 1
3.	DAP Submission	Week 4
4.	DAP Approval	Week 6
5.	Detailed Designing	Week 10
6.	Manufacturing & Procurement	Week 16
7.	Inhouse Assembly	Week 20
8.	Dispatch	Week 21

**Above plan is tentative, detailed plan will be shared post order confirmation.*

21.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Amazon
3. Weekly/Fortnightly Meeting Amazon to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

22. Amazon Responsibility

22.1 Amazon Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- Amazon to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for

the installation activities (estimated at 20 kW).

- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

22.2 Responsibilities of Amazon During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

22.3 Amazon Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of AMAZON.

23.Commercial Terms

23.1Price Sheet

S. No	Component	Set	Price (INR)	
1	Linear Cross Belt Sorter	1	₹	2,88,29,302
2	Conveyor Automation	1	₹	37,70,223
3	Output Chutes	1	₹	26,79,662
4	Steel Works	1	₹	15,48,650
5	Software Packages	1	₹	11,00,000
6	Toolkit	1	₹	79,800
7	Installation & Commissioning	1	₹	18,45,382
8	Packaging & forwarding	1	₹	7,38,153
9	Project Management	1	₹	10,14,779
Grand Total (INR)			₹	4,16,05,951

The Total Price is to be understood.

- Freight - Excluded
- Taxes as applicable
- Packing and Forwarding - Included
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area is in customer scope

23.2Payment Terms

50% Advance with PO.

30% With readiness of material before dispatch.

20% After successful installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer*

24. Warranty Period

Falcon's offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

25. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Server System**
- **Mobile Carts**
- **Construction Power**
- **Steel Works- If not specified**
- **Collection Trolleys below chutes**
- **PC for SCADA.**
- **Stress free mats**
- **Insulation mats**
- **Fans at chutes & inducts.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.